

AI Job Market Analysis using SQL & Power BI

1. INTRODUCTION

Artificial Intelligence (AI) has become one of the most influential technologies in the modern era. Industries such as healthcare, finance, retail, and manufacturing are increasingly adopting AI solutions. As a result, the demand for AI professionals has grown rapidly. However, understanding job market trends, required skills, and salary expectations can be challenging due to the large volume of job data available. This project addresses this challenge by analyzing AI job market data using SQL and Power BI.

2. PROBLEM STATEMENT

Job seekers often struggle to understand which AI skills are in demand, which roles offer higher salaries, and which locations or companies provide better opportunities. Organizations also face challenges in analyzing hiring trends due to unstructured and scattered job data. This project aims to convert raw job listings into meaningful insights through structured analysis and visualization.

3. OBJECTIVES OF THE PROJECT

- To analyze the overall AI job market
 - To study salary distribution across job roles and experience levels
 - To identify in-demand and high-paying skills
 - To analyze company-wise and location-wise hiring trends
 - To create interactive dashboards for decision-making
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4. DATASET DESCRIPTION

The dataset used in this project consists of AI-related job listings. It includes the following attributes:

- Job Title
- Company Name
- Location
- Salary
- Experience Level
- Employment Type
- Required Skills

5. TOOLS & TECHNOLOGIES USED

- **SQL:** Used for querying, filtering, and aggregating job data
 - **Power BI:** Used for creating interactive dashboards
 - **Power Query:** Used for data cleaning and transformation
 - **DAX:** Used for calculated measures and KPIs
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6. DATA PREPROCESSING & CLEANING

Data preprocessing was an essential step in this project. The following steps were performed:

- Removal of duplicate records
- Handling missing salary values
- Standardizing salary formats
- Splitting skill columns into individual values
- Creating calculated columns and measures

These steps ensured that the data was clean and ready for analysis.

7. SQL ANALYSIS

SQL queries were written to extract insights such as:

- Total number of AI jobs
- Average salary by job role
- Top hiring companies
- Location-wise job distribution
- Skill-wise demand

SQL helped in efficient data aggregation and filtering before visualization.

8. DASHBOARD 1: JOB MARKET OVERVIEW

This dashboard provides a high-level overview of the AI job market. It includes:

- Total number of job openings
- Distribution of jobs by experience level
- Employment type analysis (Full-time, Contract, Remote)

This dashboard helps understand the overall structure of the AI job market.

9. DASHBOARD 2: SALARY INSIGHTS

The Salary Insights dashboard focuses on compensation trends. It includes:

- Average salary by job role
- Salary comparison by experience level
- Top 10 highest paying jobs
- Highest and lowest salary indicators

This dashboard helps job seekers understand salary expectations.

10. DASHBOARD 3: SKILLS & DEMAND ANALYSIS

This dashboard analyzes skill requirements in the AI job market. It includes:

- Most in-demand AI skills
- Skill distribution using donut charts
- Skill complexity analysis
- Relationship between skills and salary

This dashboard highlights the importance of multi-skilled professionals.

11. DASHBOARD 4: COMPANY & LOCATION INSIGHTS

This dashboard focuses on hiring organizations and regions. It includes:

- Top hiring companies
- Job distribution by location
- Salary comparison by region
- Company-wise salary trends

It helps identify major hiring hubs and organizations.

12. RESULTS & KEY INSIGHTS

- Multi-skilled AI professionals earn higher salaries
- Certain locations dominate AI job opportunities
- Advanced AI skills are in high demand
- Large companies offer better salary packages

13. CONCLUSION

This project demonstrates how SQL and Power BI can be effectively used to analyze real-world job market data. The insights derived from this analysis can help job seekers, recruiters, and organizations make informed decisions. The dashboards provide a clear and interactive way to understand complex job market trends.

14. FUTURE SCOPE

- Integration of real-time job portal data
- Predictive salary analysis using machine learning
- Time-based trend analysis
- Industry-specific AI job analysis